

Kien Pham

✉ ki3n98@gmail.com ☎ 714 592 6510 📍 Westminster, CA 🌐 Kien Pham 🔄 ki3n98 📁 Portfolio

Education

Bachelors of Science in Computer Science, WIP
California State University, Long Beach 📄

08/2024 - 12/2026
Long Beach, CA

Internship

Thomson Reuters Special Services
Data Science Intern

06/2026 - Present
Remote

- Explore and build 80+ embedding/model combinations to detect victims of juvenile trafficking in escort ads, lifting recall from .620 to 0.785 (+26%).
- Run SHAP, feature importance, and attention-based explainability to identify the linguistic signals driving predictions.
- Validate model robustness on out-of-year data and build drift-detection tooling to catch vocabulary shift over time.

Projects

VeriFace

06/2026

2026 CECS Senior Design Project Award for Outstanding Achievement in Applied AI & ML

- Full-stack AI-powered attendance system that uses facial recognition, liveness detection, and real-time WebSockets to automate and secure check-ins for classrooms and events.
- Implemented an end-to-end attendance platform using FaceNet (InceptionResnetV1) and MTCNN to generate 512-dim L2-normalized face embeddings stored in PostgreSQL.
- Designed a real-time attendance feed using WebSocket connections (FastAPI + custom ConnectionManager), broadcasting live check-in events to all clients watching an active session.
- Built a layered REST API (routers → services → repositories) with JWT auth, bcrypt hashing, RBAC (owner/admin/member), and an audit log using FastAPI and SQLAlchemy 2.

Vision-Based Distracted Driving Detection 📄

02/2026

Sixth Annual Computer Science Conference for CSU Undergraduates

- Developed a vision-based driver distraction detection system and benchmarked Faster R-CNN, SSD, and YOLOv11 on the Roboflow Mobile Detection dataset, identifying YOLOv11 as the optimal model (mAP = 0.604, 86.1 FPS) for real-time deployment.
- Designed a controlled experimental framework to compare two-stage and single-stage object detectors using mAP and FPS, determining the best accuracy-latency trade-off for safety-critical driver monitoring systems.
- Conducted quantitative performance analysis across architectures, delivering deployment guidance for real-time, latency-sensitive object detection applications.

911 Operator Assistance 📄

10/2025

Marina Hack 5.0 - Best Overall

- Won Best Overall for an AI copilot that transcribes, analyzes, and summarizes emergency calls so operators can focus on decision-making and coordination instead of note-taking.
- Built a live pipeline: Whisper TTS → Gemini 2.5 Flash entity extraction (location/emergency type) → Google Maps API geocoding, enabling precise, clickable coordinates in the Next.js dashboard.
- Integrated Twilio Programmable Voice to receive real phone calls and drive realistic call flows; streamed audio to models and surfaced live insights to the operator UI.

Skills

Machine Learning & AI: Computer Vision, NLP, fine-tuning, Prompt Engineering, LangChain, Multi-agent orchestration
Frameworks & Tools: TensorFlow, PyTorch, Scikit-Learn, Django, FastAPI, Next.js, Git/GitHub, Docker
Data: PostgreSQL, Matplotlib, data collection, preprocessing, data evaluation, data labeling
Programming Language: Python, C++, C#